Sir Syed CASE Institute of Technology

Islamabad

WEB ENGINEERING

Lab 03

CSS Fundamentals:
Styling, Layout and Animation

Lab Report

SUBMITTED BY

Muhammad Hamza

ROLL NUMBER

2330-0068

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CSS exercises, browser results and submission evidence

Selectors, colours and text

Tasks 1 and 2 | Stylesheet sources and text properties

TASK QUESTION

Create a blue div with id box, red h1 to h3 headings using the shared class heading, and buttons coloured green from style.css, blue from a style tag and pink inline. Add a centred, capitalised heading and use Times New Roman throughout the document.

The id selector colours the box blue. One class colours all three headings red. The three buttons demonstrate external, internal and inline CSS. The document inherits Times New Roman, and the text example uses CSS capitalisation and centre alignment.

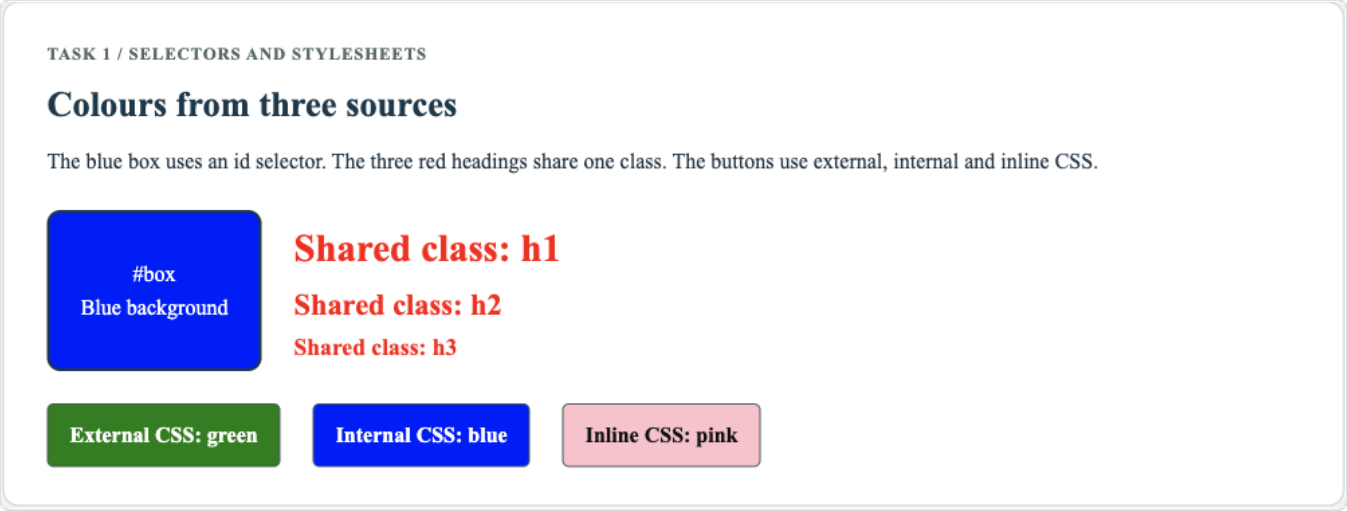


Figure 1. Blue box, shared red headings and the three stylesheet sources.

KEY CSS RULES

```
body { font-family: "Times New Roman", Times, serif; }
#box { background-color: blue; }
.heading { color: red; }
.external-button { background-color: green; }
.capitalised-heading {
  text-align: center;
  text-transform: capitalize;
}
```

The blue button rule is inside a style element in the head. The pink button has an inline style attribute. The lowercase heading source is displayed with each word capitalised.

Pixels, em and rem

Task 3 | Nested text sizes

TASK QUESTION

Create an outer div containing an inner div. Set the outer text to 25px and the inner text to 10px. Change the sizes to em and rem and explain the difference.

Three versions of the nested boxes are shown together so the sizes can be compared. The root font size is 16px. The browser confirms that each outer box resolves to 25px and each inner box to 10px.

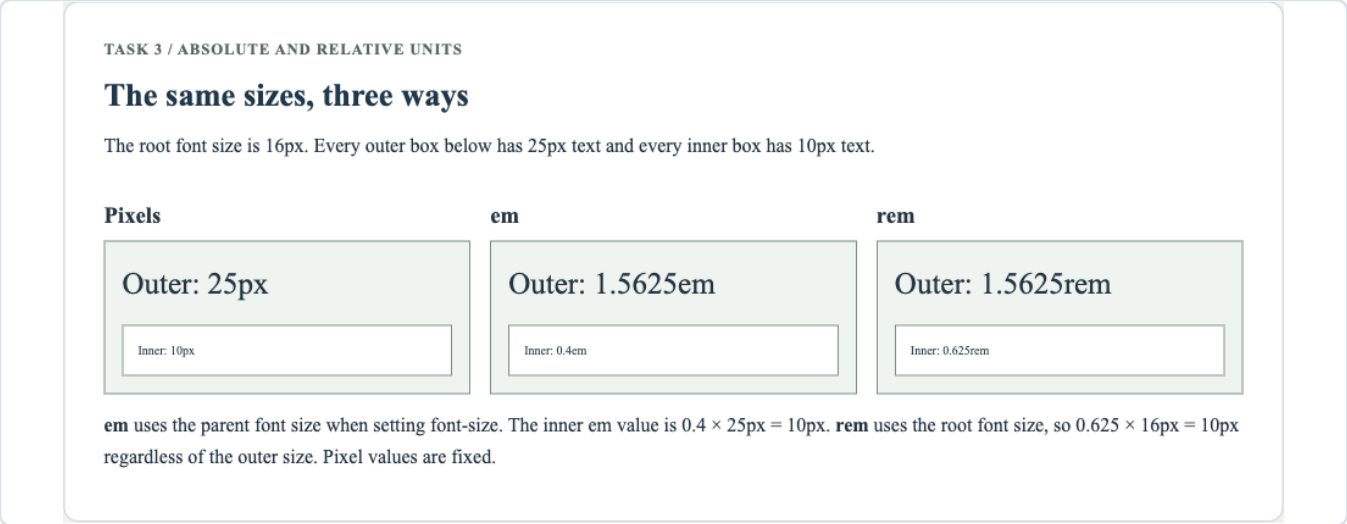


Figure 2. The px, em and rem examples produce matching text sizes.

Unit	Outer value	Inner value
px	25px	10px
em	1.5625em	0.4em
rem	1.5625rem	0.625rem

em: When setting font-size, em uses the parent font size. The outer value is 1.5625 × 16px = 25px. The inner value is 0.4 × 25px = 10px.

rem: Both values use the root size. The outer value is 1.5625 × 16px = 25px and the inner value is 0.625 × 16px = 10px. Changing the outer size does not change the rem reference.

px: The declared size is fixed and does not depend on the parent font size.

Navbar and box model

Tasks 4 and 5 | Page layout and Flexbox

TASK QUESTION

Build a 60px navbar with background #0f1111, a 25px logo in #f08804, three middle links and a search box with an orange button. Use Flexbox in one row. Add a header, footer and content area with three bordered 100px boxes at opacity 0.5; make the last a circle.

The header navbar spaces its three groups evenly. The content area contains the three boxes. The browser verifies their 100px outer dimensions, 2px borders and 0.5 opacity.



Figure 3. The navbar with its logo, three links and search form.

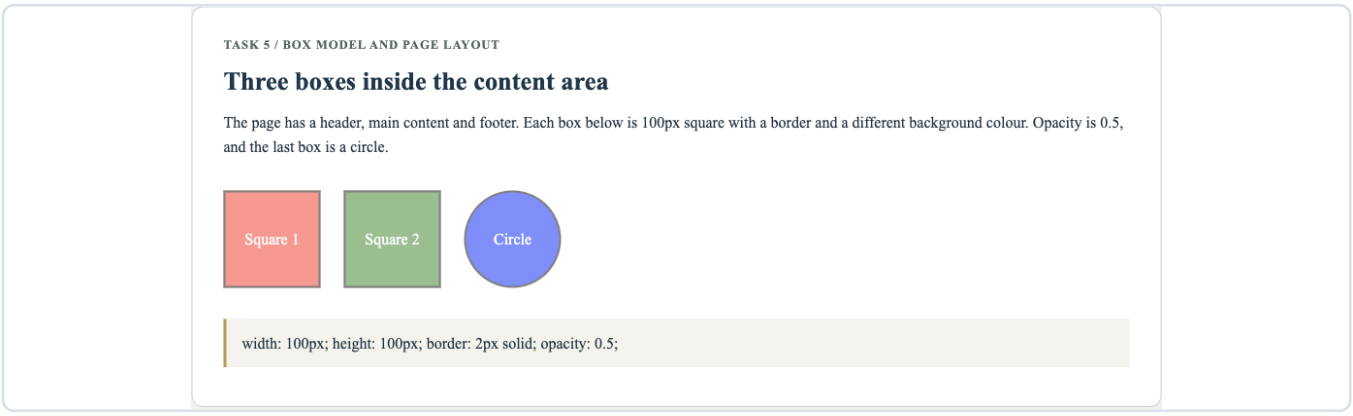


Figure 4. Three boxes with different colours; the last has a circular border.

LAYOUT RULES

```
.navbar {
  display: flex; height: 60px;
  justify-content: space-evenly; align-items: center;
  flex-wrap: nowrap; background-color: #0f1111;
}
.layout-box {
  width: 100px; height: 100px;
  border: 2px solid black; opacity: 0.5;
}
.box-three { border-radius: 50%; }
```

The universal box-sizing rule includes padding and borders within the 100px size. On narrow screens, the navbar scrolls horizontally while keeping its required single row.

Fixed nature banner

Task 6 | Background image and stacking

TASK QUESTION

Between two paragraphs, add a Love Nature div with a background image. Fix it to the right edge so it does not move on scroll, and use z-index to keep it above the page content.

The banner uses an original SVG nature illustration as its background. It is fixed to the right edge with z-index 10. The two screenshots show the page before and after a 280px scroll.

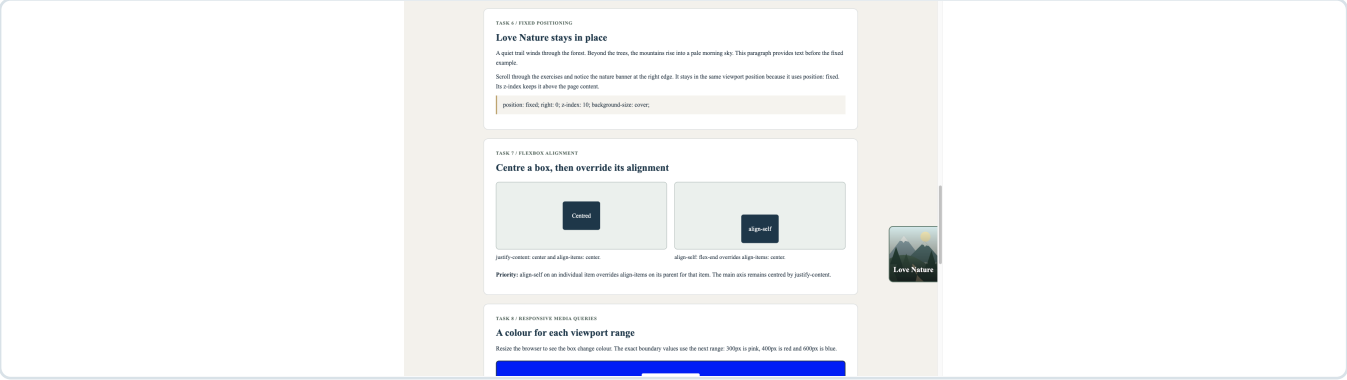


Figure 5. The banner before scrolling, at viewport coordinates x = 1295, y = 600.

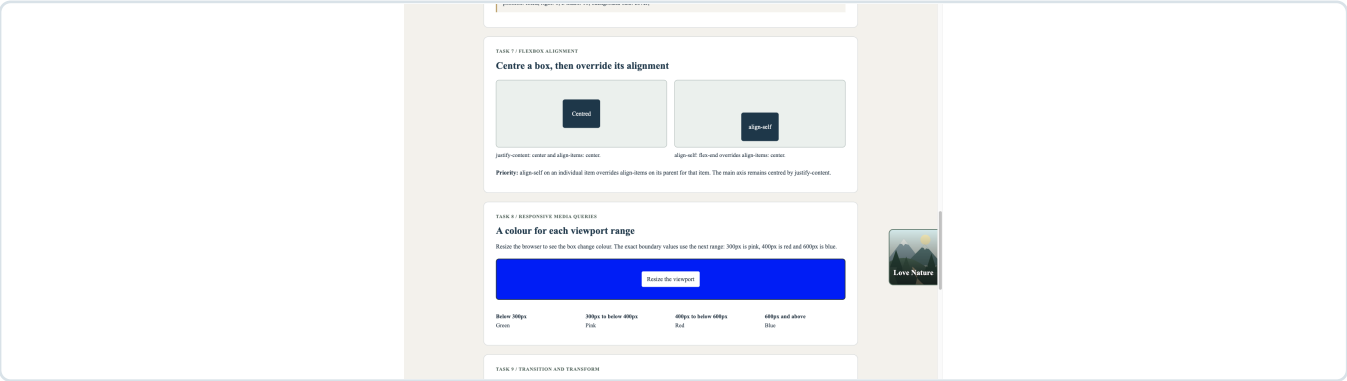


Figure 6. The content moves after scrolling; the banner stays at the same coordinates.

```
.nature-banner {
  position: fixed; right: 0; top: 60vh; z-index: 10;
  background-image: url("nature.svg");
  background-size: cover;
}
```

The banner is smaller on narrow screens so the fixed example takes less space.

Flexbox alignment

Task 7 | Centre alignment and item overrides

TASK QUESTION

Use Flexbox to centre one div inside another. Explain which takes priority: align-items or align-self.

The first parent centres its child on both axes. The second parent has the same rules, but its child uses align-self: flex-end. The child moves to the end of the cross axis while staying horizontally centred.

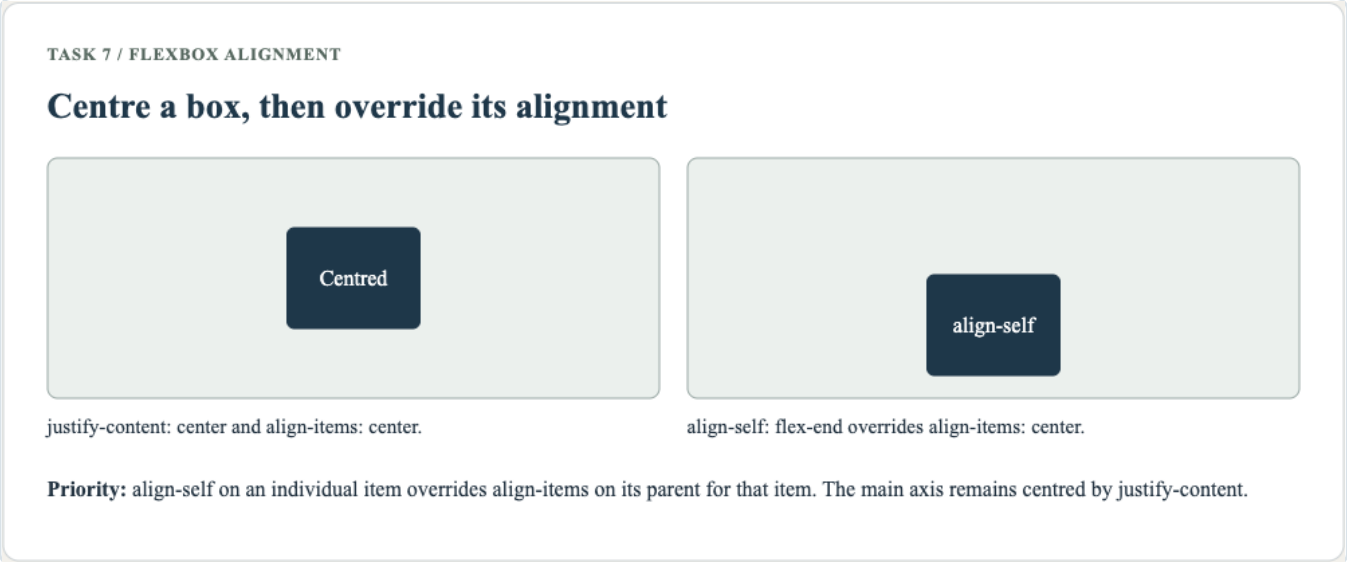


Figure 7. Centre alignment on the left and the item override on the right.

PARENT AND CHILD RULES

```
.centering-parent {
  display: flex;
  justify-content: center;
  align-items: center;
}
.override-child { align-self: flex-end; }
```

Priority: align-self on the item takes priority over align-items on the parent for that item. justify-content controls the main axis and is unchanged by this override.

Browser check: The first child has the same horizontal and vertical centre as its parent. The overridden item ends 17px from the outer bottom edge, accounting for the parent padding and border.

Responsive colour changes

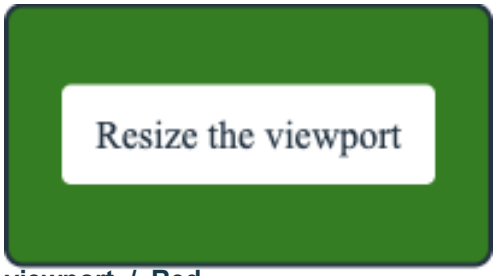
Task 8 | Media queries and boundary checks

TASK QUESTION

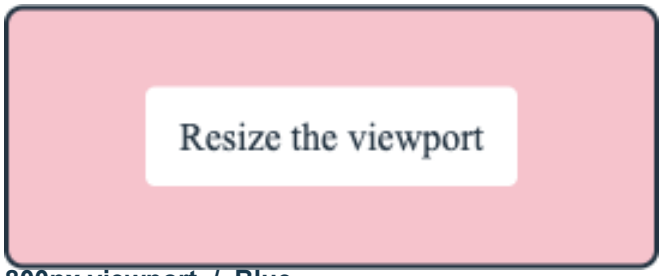
Change a div colour with media queries: green below 300px, pink from 300px to 400px, red from 400px to 600px and blue above 600px.

The base colour is green. Later media queries replace it as the width increases. Exact boundary widths use the next range: 300px is pink, 400px is red and 600px is blue.

280px viewport / Green



350px viewport / Pink



500px viewport / Red



800px viewport / Blue



Figure 8. Actual browser captures at four viewport widths.

MEDIA QUERIES

```
.responsive-box { background-color: green; }
@media (min-width: 300px) {
  .responsive-box { background-color: pink; }
}
@media (min-width: 400px) {
  .responsive-box { background-color: red; }
}
@media (min-width: 600px) {
  .responsive-box { background-color: blue; }
}
```

Verified widths: 280, 300, 350, 399, 400, 500, 599, 600 and 800px. All colours matched the expected range. The page had no horizontal overflow at those widths.

Transition and transform

Task 9 | Button hover effect

TASK QUESTION

Add a hover transition to a button that grows its font size and rotates it slightly with transform.

The button starts at 16px. Hovering changes it to 24px and rotates it by five degrees. Both properties transition over 0.4 seconds. Keyboard focus uses the same effect.

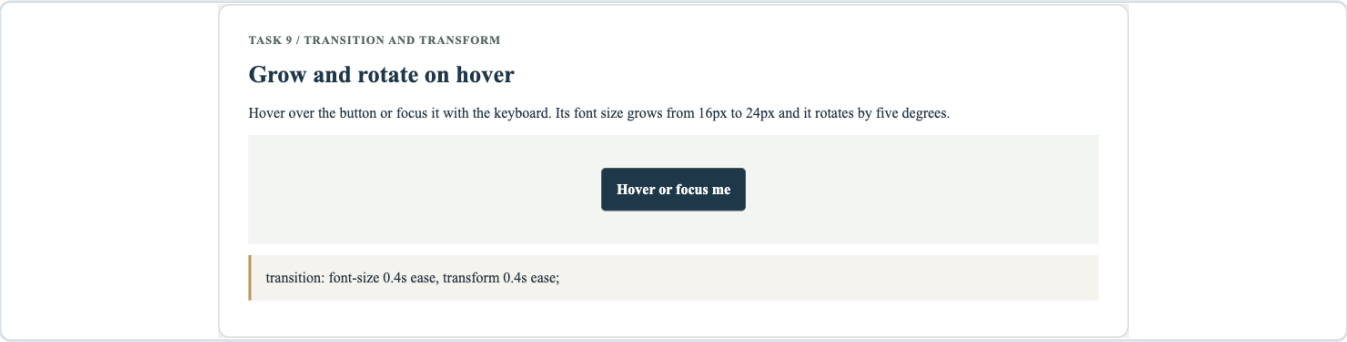


Figure 9. The button before hover, with 16px text and no rotation.

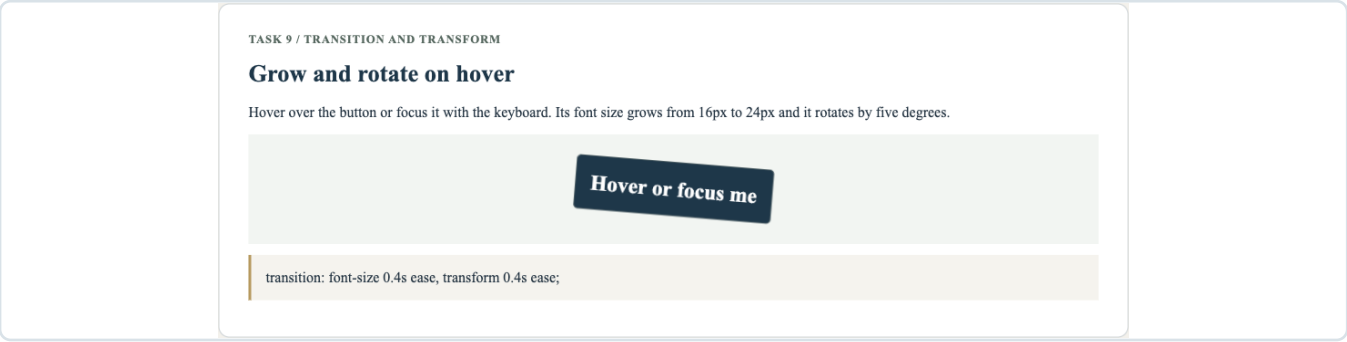


Figure 10. The button after hover, with 24px text and a five degree rotation.

```
.transition-button {
  font-size: 16px;
  transition: font-size 0.4s ease, transform 0.4s ease;
}
.transition-button:hover {
  font-size: 24px;
  transform: rotate(5deg);
}
```

The browser checked the final font size and transform matrix after the transition. Reduced motion disables rotation and the animated transition.

Circular loader

Task 10 | Keyframe animation

TASK QUESTION

Create a circular loader with a thick border coloured on one side only. Spin it from 0 to 360 degrees using a keyframe animation that repeats infinitely.

The loader is a 72px circle with an 8px border. Only the top border is coloured. The spin animation runs continuously with a one second duration and linear timing.

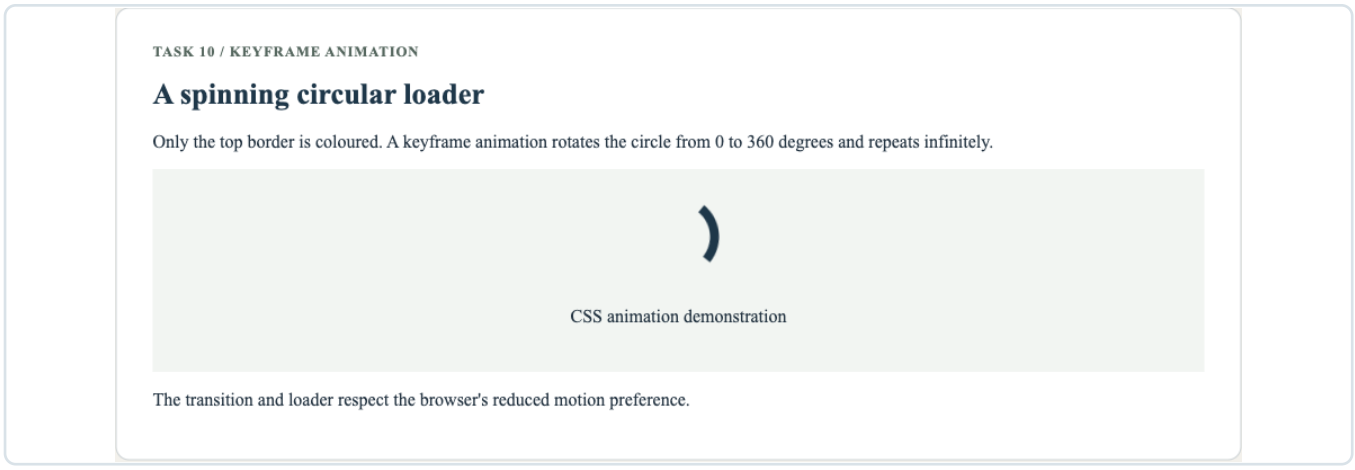


Figure 11. The circular loader in the browser.

LOADER AND KEYFRAMES

```
.loader {
  width: 72px; height: 72px;
  border: 8px solid transparent;
  border-top-color: #17374b;
  border-radius: 50%;
  animation: spin 1s linear infinite;
}
@keyframes spin {
  from { transform: rotate(0deg); }
  to { transform: rotate(360deg); }
}
```

Browser result: The animation name is spin and the iteration count is infinite. The transform changed between two samples, confirming that the loader rotates. The other three border sides remain transparent.

The loader is decorative and hidden from assistive technology. With reduced motion enabled, the animation is disabled.

Verification and submission

Browser checks, tooling and GitHub evidence

All ten exercises were checked in Chrome using computed styles and element positions. The repository keeps the earlier labs and adds the new css-lab folder. The source files and browser evidence were committed and pushed to main.

Check	Result
CSS exercises	All ten passed
Responsive widths	Nine widths passed
Browser errors	0
Lint, formatting and existing test	Passed

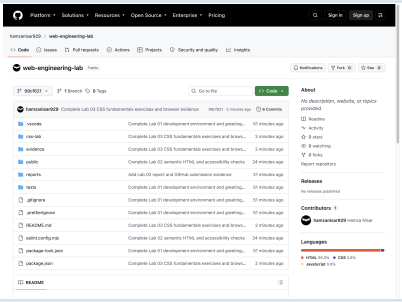


Figure 12. The GitHub repository with the css-lab folder and Lab 03 commit.

PUSH CONFIRMATION

```
To github.com:hamzanisar929/web-engineering-lab.git
   08681d4..99cf621  main -> main
```

Repository: https://github.com/hamzanisar929/web-engineering-lab

Local project: Lab 3/web-engineering-lab/css-lab
Evidence: evidence/lab03 contains screenshots, computed CSS results, viewport checks and Git output.

REFERENCE

Lab 03 CSS Fundamentals Lab Manual, SSCIT Format, Sir Syed CASE Institute of Technology, Islamabad. Task questions are restated from the supplied manual. CASE logo from the supplied Lab 01 handout. The nature background is an original SVG included with the project.